Lectures 15 – 18

Math 4741: Mathematical Analysis

Chapter 7: Renewal Theory, Semi Markov Process and M/G/1

References

Oliver Ibe – Markov Processes for Stochastic Modeling (Chapter 6)

Sheldon Ross – Introduction to Probability Models (Chapter 7)

Introduction

MRGP – Markov Regenerative Process

SMP – Semi–Markov Process

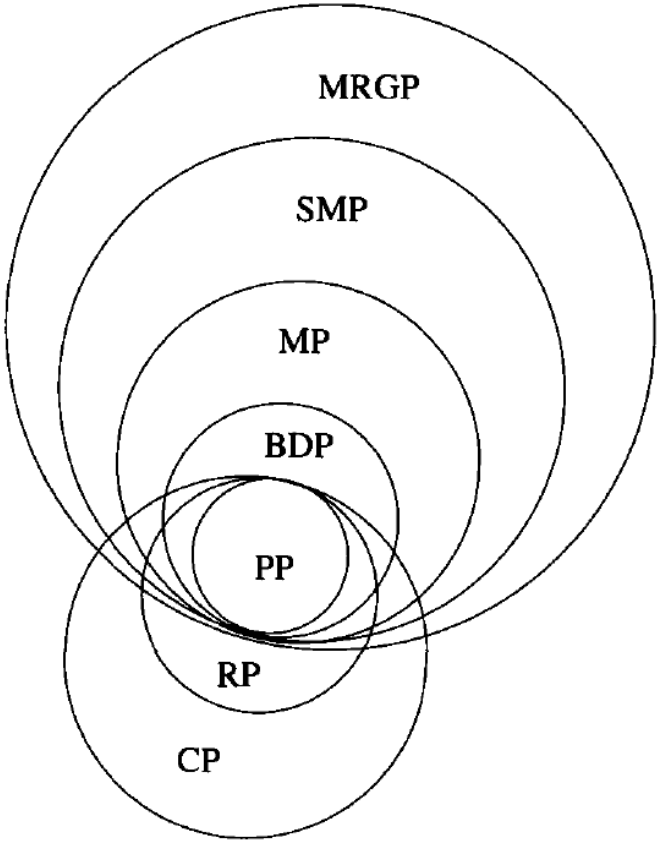
MP – Markov Process

BDP – Birth–and–death Process

PP – Poisson Process

RP – Renewal Process

CP – Counting Process



Introduction [contd.]

For a Stochastic process with discrete state space:

1. Where does the process go next?

   Transition probabilities

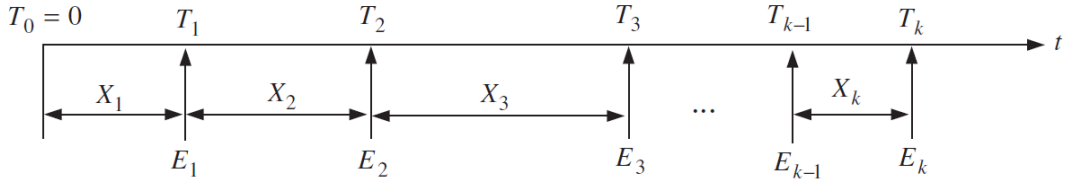
2. How long does the process stay in the current state?

   Holding time

Renewal Process

Experiment:

- A set of identical bulbs
- Use one bulb at a time
- If a bulb fails, replace it immediately



For the process $\{N(t), t \geq 0\}$,

- CP
-
- $N(t)$
-
- $\{T_n, n \geq 1\}$
-
- $X_n, n = 1, 2, \ldots$

Equivalent Events in $\{N(t), t \geq 0\}$

- $\{N(t) \geq n\}$ vs $\{T_n \leq t\}$
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- $\{N(t) < n\}$ vs $\{T_n > t\}$

Probability distribution of $N(t)$

$$P_{N(t)} = \mathrm{P}[\mathrm{N(t)} \ = \ \mathrm{t}]$$

The Renewal Function

$H(t) \equiv$ Expected number of renewals

Example 7.1: Consider a renewal process, where interarrival times $X_n, n = 1,2, \ldots$ are IID geometric (p). $T_1 = X_1$ and represents # trials to get the first renewal.

$T_n \equiv$ no of trials to get the n-th renewal

$P[T_n = k] =$

$P[N_k = n] =$

$P[N_k = n] =$

Can also be calculated by conditioning on $\{T_n = y\}$

$P[N_k = n] =$

$P[N(t) = n] =$

Example 7.2

The Renewal Equation

$\{N(t), t \geq 0\}$ is a renewal process and T_1 first renewal time with pdf $f_T(t)$

$H(t) =$

Elementary Renewal Theorem

If $H(t)$ denotes the expected number of renewal by t, then

$$\lim_{t\to\infty}\left\{\frac{H(t)}{t}\right\} =$$

$$\lim_{t\to\infty}\left\{\frac{N(t)}{t}\right\} =$$

$$\lim_{t\to\infty}\left\{\frac{EN(t)]}{t}\right\} =$$

Renewal Reward Process

If in a renewal process a reward is associated with each renewal

Y_n is reward in the n-th renewal event

Relation to the compound Poisson process

Reward Renewal Theorem

If $E[Y_1] < \infty$ and $E[X_1] < \infty$ then

$$\lim_{t \to \infty} \left\{ \frac{Y(t)}{t} \right\} =$$

Cycle: Every time a renewal starts a cycle is completed

Long-term Average reward is

Semi Markov Process

Consider a finite state stochastic Process where

- Prob of transition to next state j depends on the current state i
- Holding time in state i has an arbitrary distribution

$S =$

$X_n =$

$T_n =$

Semi Markov Process

2-dimensional SP $\{\{X_n, T_n\}, n = 0,1, \dots\}$ is an SMP if

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-

Discrete-time and Continuous-Time Semi Markov Processes

Process	Transition to j from i	Time to transition	Holding-time Distribution	Markovian

Discrete-Time Semi Markov Process

Two Components

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Probability Models

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-
-
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Embedded Markov Chain

- Only consider the transitions between states and ignore the times between transitions

Limiting Probability vector

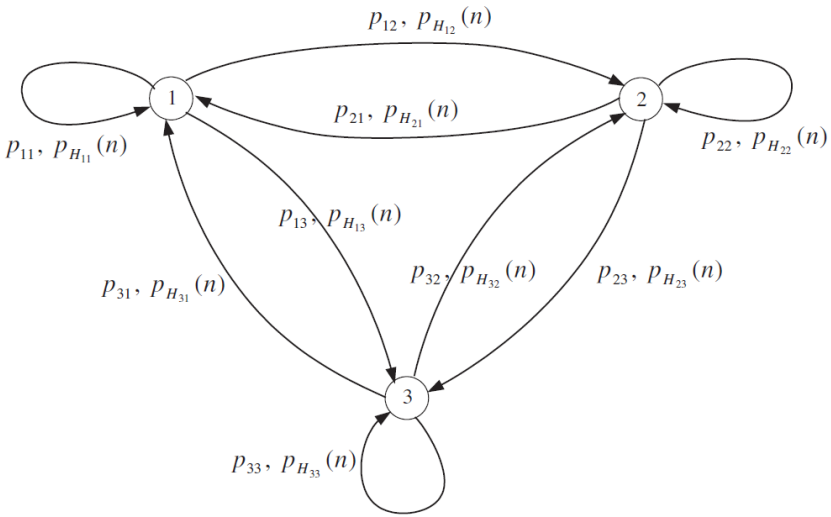
Holding Times

Each time the process visits state i

- It chooses the next state j
- It spends a random time depending on j

Distribution of H_{ij}

Distribution of H_i


$p_{12}, p_{H_{12}}(n)$
$p_{21}, p_{H_{21}}(n)$
$p_{11}, p_{H_{11}}(n)$
$p_{22}, p_{H_{22}}(n)$
$p_{13}, p_{H_{13}}(n)$
$p_{31}, p_{H_{31}}(n)$
$p_{32}, p_{H_{32}}(n)$
$p_{23}, p_{H_{23}}(n)$
$p_{33}, p_{H_{33}}(n)$
1
2
3

Long Term behavior of the process

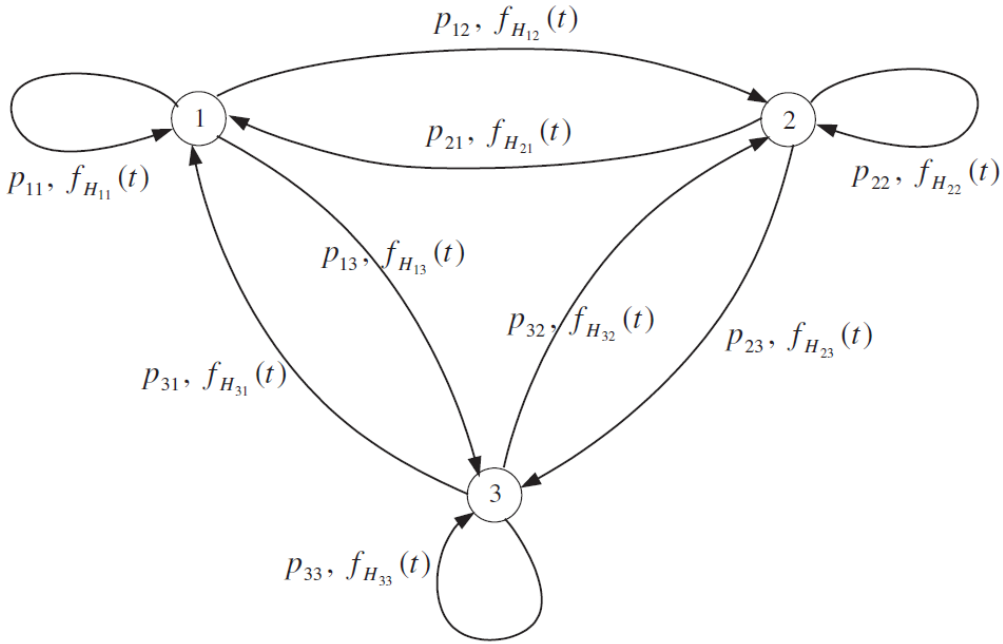
- Long term Prob. of visiting i
- Long term fraction of time staying in i

Relation to the Renewal Reward Process

Example:

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 1 & 0 & 0 \end{bmatrix}$$

Continuous-Time SMP



$\{N(t), t \geq 0\}$ $\{T_n, n \geq 0\}$

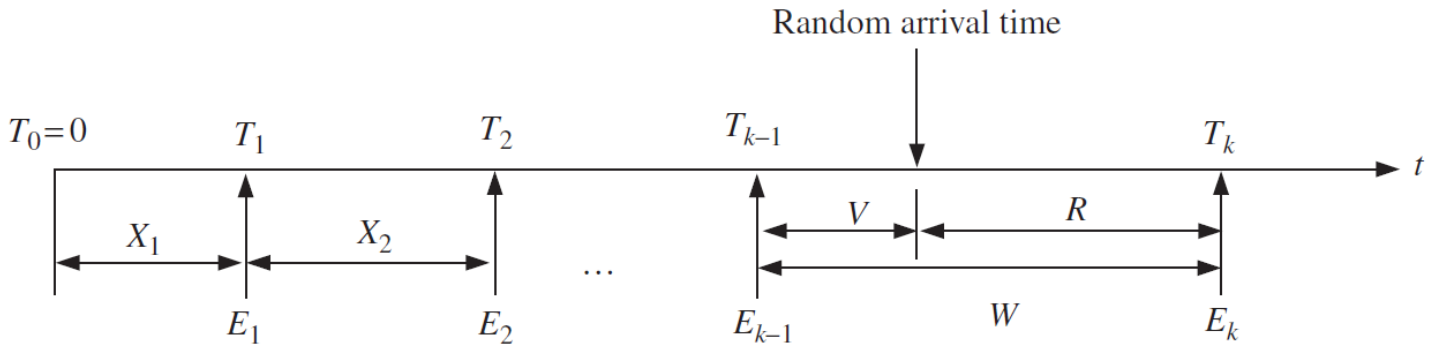
$X_n = T_n - T_{n-1}$

$T_k \equiv$ Random arrival of the k-th customer

$R \equiv$ RV representing the remaining service time of the i-th customer

= residual lifetime of the renewal process

Random Incidence and Residual Time



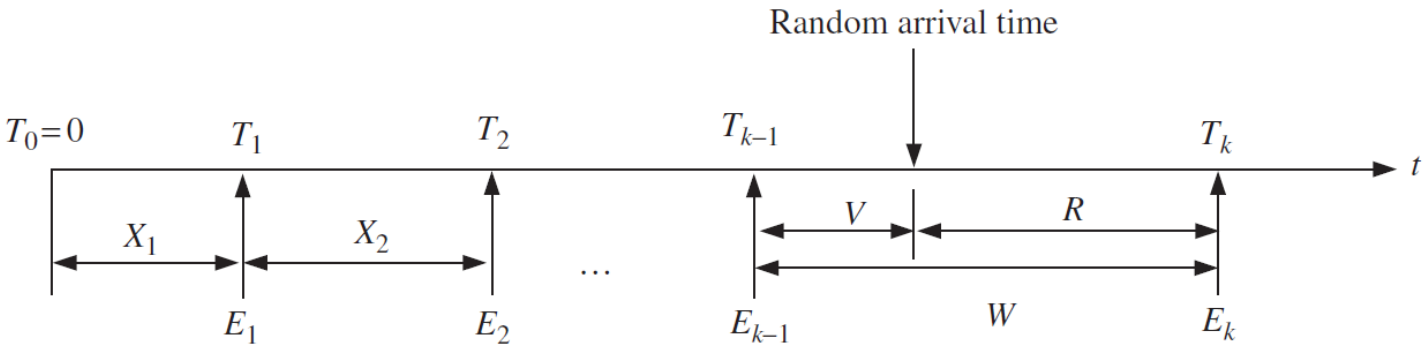
$f_X(x)$ $f_W(w)$ $f_R(r)$

Probability that the random arrival occurs between

Distribution of W, $f_W(w)$

Example: Poisson Process

Random Incidence Paradox


Random arrival time
$T_0=0$
T_1
T_2
T_{k-1}
T_k
t
X_1
X_2
V
R
E_1
E_2
E_{k-1}
W
E_k

Distribution of R, $f_R(r) \rightarrow$ Conditional PDF of R, given $W = w$

Joint distribution of R and W

Marginal PDF of R, $f_R(r)$

Expected value of R

Example Poisson Process

$M/G/1$ Queueing System

Single server queueing system

- Arrival
- Service
- Number of servers
- Population size
- System capacity
- Queueing discipline

Service Time distribution $X \sim$

State of the queue $N(t)$

Is $\{N(t), t \geq 0\}$ a Poisson Process?

Next state depends on

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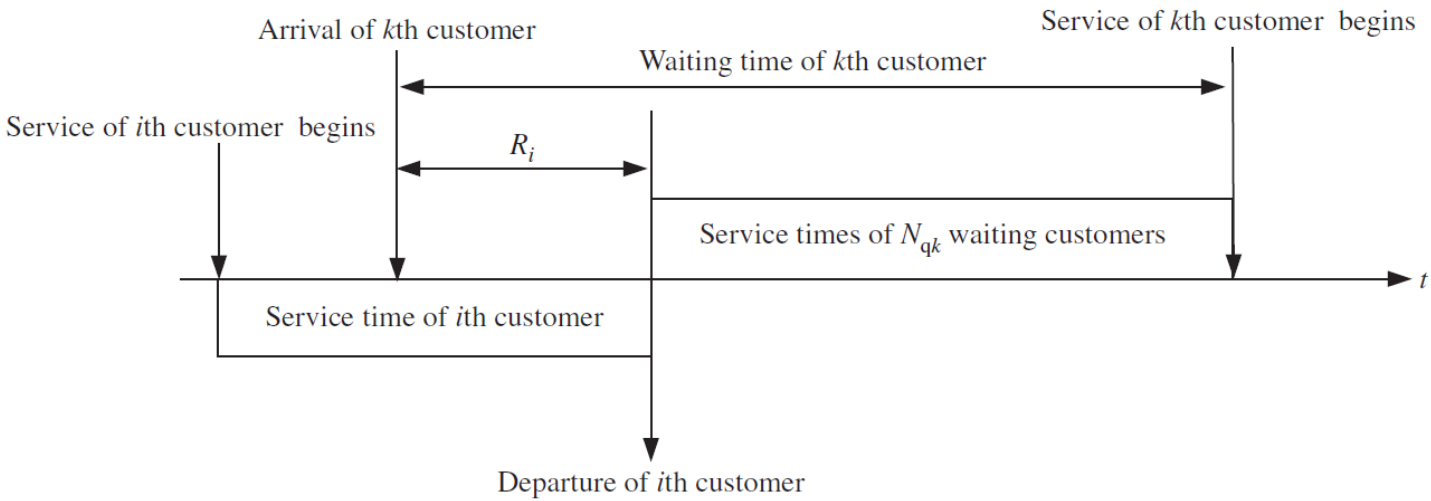
-

Simple Approximation


Arrival of kth customer
Service of kth customer begins
Waiting time of kth customer
Service of ith customer begins
R_i
Service times of N_{qk} waiting customers
t
Service time of ith customer
Departure of ith customer

Residual service time of the i-th customer

Queue size at the arrival time of the k-th customer

Total number of customers when k-th customer arrived

Then the waiting time of the k-th customer

Expected waiting time of the k-th customer

Because the experience of the k-th customer is typical, mean waiting time in an M/G/1 system

Expected number of customers in the system (Pollaczek-Khinchin formula)